

Chapter 20

Upwind Schemes for the Euler Equations

The schemes discussed in Chapters 17 and 18 are based on central space discretizations and have a symmetry with respect to a change in sign of the Jacobian eigenvalues which does not distinguish upstream from downstream influences. Hence the physical propagation of perturbations along characteristics, typical of hyperbolic equations, is not considered in the definition of the numerical model.

The family of upwind schemes, whose origin may be taken back to Courant, Isaacson and Reeves (1952), is directed towards an introduction of the physical properties of the flow equations into the discretized formulation and has led to the family of techniques known as upwinding, covering a variety of approaches, such as flux vector splitting, flux difference splitting and various 'flux controlling' methods.

Other schemes based on characteristic formulations, such as the λ scheme of Moretti (1979), also rely on the physical propagation information contained in the equations. They are, however, non-conservative and require some shock fitting in the presence of discontinuities.

As seen earlier, all second-order central schemes generate oscillations in the vicinity of discontinuities, which have to be damped by the addition of artificial dissipation terms. Constructing schemes which take into account the essential physical properties of the equations aim at preventing the creation of unwanted oscillations.

In smooth regions of the flow, where the flow variables can be considered as continuous, the central schemes based on Taylor series expansions can be applied with any order of accuracy. This will be the case even for supersonic flows, the apparent contradiction between the physical one-way propagation of waves and the symmetrical central differenced schemes which are direction independent, being resolved by considering the analytic continuation properties of smooth functions. The validity of Taylor series expansions expresses, indeed, a most remarkable property of continuous functions, namely that it suffices to know the value of a continuous function in a single point, together with a sufficiently high number of derivatives in that same point, to be able to reconstruct the function in an increasingly large domain around that point. In

the limit, the complete local knowledge of a function, that is the knowledge of its value and all its derivatives in a single point, is equivalent to the knowledge of the function everywhere. However, as soon as discontinuities appear this information is destroyed and more physical input is required in order to resolve the non-linear behaviour.

The introduction of physical properties in the discretization process of the Euler equations can be done at different levels.

The first level introduces only information on the sign of the eigenvalues, whereby the flux terms are split and discretized directionally according to the sign of the associated propagation speeds. This leads to the *flux vector splitting* methods and will be described in Sections 20.2 to 20.4.

A higher level of introduction of physical properties into the definition of the scheme can, however, be defined, following the very remarkable scheme of Godunov (1959). In Godunov's method, the conservative variables are considered as piecewise constant over the mesh cells at each time step and the time evolution is determined by the exact solution of the Riemann (shock tube) problem at the inter-cell boundaries. Hence, properties derived from the exact local solution of the Euler equations are introduced in the discretization. This approach has been extended to higher orders, as well as to variants, whereby the local Riemann problem is only approximately solved through approximate Riemann solvers. They are referred to sometimes as *flux difference splitting* methods and we will refer to the family of methods which call on exact or approximate local properties of basic solutions to the Euler equations as *Godunov-type* methods.

Section 20.1 introduces the principles of the upwind schemes, while the basic concepts and properties of flux vector splitting are discussed in Section 20.2. The first-order discretization techniques applied to one-dimensional flux vector splitting are presented in Section 20.3 and extended to multi-dimensions in Section 20.4. Section 20.5 presents the basic first-order Godunov scheme and the approximate Riemann solvers of Roe and Osher.

The schemes presented in this chapter are restricted to first-order accuracy, while higher-order upwind schemes will be discussed in Chapter 21.

20.1 THE BASIC PRINCIPLES OF UPWIND SCHEMES

The original scheme of Courant *et al.* (1952) was based on the characteristic form of the equations $u_t + au_x = 0$ and a discretization depending on the sign of the eigenvalue a .

With a first-order forward difference in time, namely the explicit Euler method, it has been noted that the central difference of u_x leads to an unstable scheme. However, with a one-sided differencing the following scheme can be considered for $a > 0$:

$$u_i^{n+1} = u_i^n - \sigma(u_i^n - u_{i-1}^n) \quad (20.1.1)$$

Its amplification function is

$$G = 1 - 2\sigma \sin^2 \frac{\phi}{2} - I\sigma \sin \phi \quad (20.1.2)$$

which is represented by a circle in the complex G plane, with centre at $(1 - \sigma)$ on the real axis and radius σ , as seen in Chapter 8 in Volume 1. Hence scheme (20.1.1) will be stable for values of the Courant number σ between zero and one:

$$0 \leq \sigma \leq 1 \quad (20.1.3)$$

but unstable for negative characteristic speeds. The truncation error ε_T is

$$\varepsilon_T = \frac{a \Delta x}{2} (1 - \sigma) u_{xx} \quad (20.1.4)$$

showing that the scheme is only first-order accurate in space and time and that the equivalent equation has a dissipative term with a numerical viscosity coefficient equal to $a \Delta x (1 - \sigma)/2$.

It is to be observed that this viscosity term vanishes for $a = 0$, that is when the characteristic eigenvalues pass through zero, as will be the case in stagnation regions and at sonic transitions. Comparing with the numerical viscosity introduced by the Lax–Friedrichs scheme, which is also first-order accurate, it is seen from equation (17.1.8) that in this latter scheme the numerical dissipation never vanishes and is higher than (20.1.4) by a factor equal to $(\sigma + 1)/\sigma$.

Hence, the upwind scheme may be expected to lead to a better representation of discontinuities. This is clearly seen on Figure 20.1.1, displaying the solutions to Burgers equation for a moving discontinuity at $\sigma = 0.5$ after 80 time steps. This property of vanishing dissipation at sonic transitions is needed in order to be able to resolve sharp discontinuities, but can also lead to non-physical shocks.

Notice the odd–even oscillations of the Lax–Friedrichs scheme.

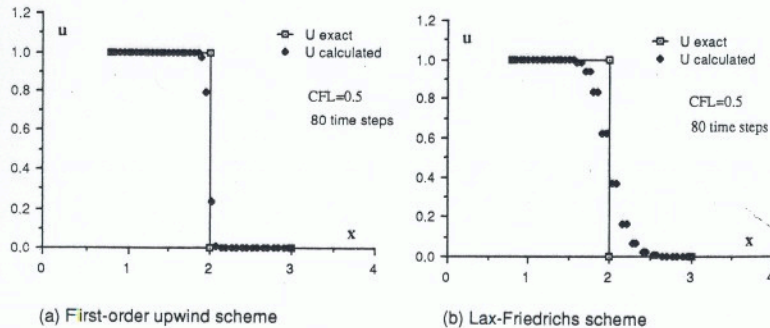


Figure 20.1.1 Solution to Burgers equation for a moving discontinuity

For negative propagation speeds, $a < 0$, the following one-sided scheme is stable:

$$u_i^{n+1} - u_i^n = -\sigma(u_{i+1}^n - u_i^n) \quad (20.1.5)$$

The amplification matrix is

$$G = 1 + 2\sigma \sin^2 \frac{\phi}{2} - I\sigma \sin \phi \quad (20.1.6)$$

and is lower than one in modulus for

$$-1 \leq \sigma \leq 0 \quad (20.1.7)$$

This shows that an upwind scheme cannot be simultaneously stable for both positive and negative eigenvalues. Steger and Warming (1981) give arguments to support this property for any non-symmetrical upwind sided scheme.

Schemes (20.1.1) and (20.1.5) are called *upwind schemes*. They apply a discretization that depends on the propagation direction of the wave or on the sign of the convection velocity a .

This also affects the boundary conditions. Scheme (20.1.1), valid for $a > 0$, will be solved by prescribing a physical boundary condition at the left side, for $i = 1$, and no numerical condition is required at the downstream end of the domain. The reverse applies to the scheme (20.1.5) which will be solved by sweeping the mesh from the downstream end, where a physical boundary condition will be applied, and no condition is necessary at $i = 1$. This is of course in full agreement with physical propagation phenomena.

A visual representation is given in Figure 20.1.2. The points involved in the discretization are always on the side of the intersection of the characteristic with the x axis; P_+ for $a > 0$ and P_- for $a < 0$.

The solution in point i at $t = (n + 1)\Delta t$ will only be influenced by the information at $t = n\Delta t$ which can be transported by the characteristic. The CFL condition expresses that all of the information physically influencing point P , which is contained in the physical domain of dependence (P_- , P , P_+), should be allowed to influence numerically the solution in P . This has already been discussed in Chapter 8 in Volume 1.

When u represents a Riemann variable, we obtain the first-order λ scheme of Moretti (1979), which is then equivalent to the Courant *et al.* scheme.

The Euler equations have generally mixed sign eigenvalues and both schemes (20.1.1) and (20.1.5) can be combined in the following way. Defining positive and negative projections of the eigenvalues

$$a^+ = \max(a, 0) = \frac{1}{2}(a + |a|) \quad (20.1.8)$$

$$a^- = \min(a, 0) = \frac{1}{2}(a - |a|) \quad (20.1.9)$$

we obtain the general form for the *first-order accurate upwind scheme* written for the linearized scalar form of the Euler equations:

$$u_i^{n+1} - u_i^n = -\tau[a^+(u_i^n - u_{i-1}^n) + a^-(u_{i+1}^n - u_i^n)] \quad (20.1.10)$$

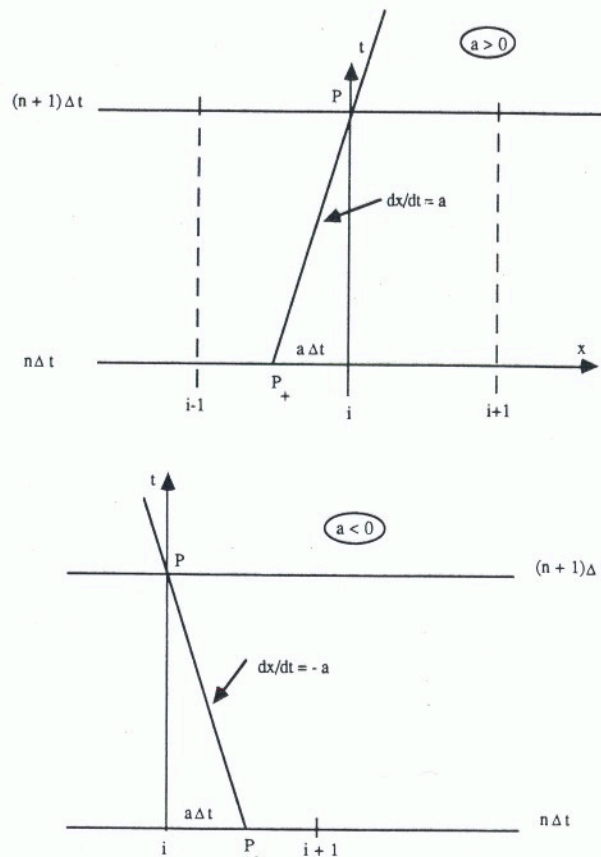


Figure 20.1.2 Characteristic properties and upwind discretization

The stability limit is now

$$|\sigma| = \tau|a| \leq 1 \quad (20.1.11)$$

Observe that a^+ is always positive (or zero), a^- is always negative (or zero) and that a problem of undeterminacy exists when $a = 0$, which has to be treated carefully.

Equation (20.1.1) can be interpreted as an integration of $du/dt = 0$ along the characteristic $dx/dt = a$ by finding $u(P_+)$ through a linear interpolation between i and $i-1$ at level n , and writing $u_P^{n+1} = u_{P_+}^n = u_i^n - \sigma(u_i^n - u_{i-1}^n)$.

It is interesting to observe at this point that the Lax-Wendroff scheme can also be interpreted in the same way by performing a quadratic symmetric

interpolation between the three points $i-1, i, i+1$ in order to estimate $u(P_+)$. Indeed, with the second-order interpolation formula

$$u(x) = u_i + \frac{u_{i+1} - u_{i-1}}{2\Delta x}(x - x_i) + \frac{u_{i+1} - 2u_i + u_{i-1}}{2\Delta x^2}(x - x_i)^2 \quad (20.1.12)$$

the Lax-Wendroff scheme is obtained from this relation at $x - x_i = -a\Delta t$

$$u_P^{n+1} = u_{P_+}^n = u_i^n - \frac{\sigma}{2}(u_{i+1}^n - u_{i-1}^n) + \frac{\sigma^2}{2}(u_{i+1}^n - 2u_i^n + u_{i-1}^n) \quad (20.1.13)$$

This shows one way of defining second-order upwind schemes, for instance by a quadratic interpolation between $i, i-1$ and $i-2$. A general presentation of this approach can be found in Yang (1986).

It is interesting to compare again this first-order scheme with the Lax-Friedrichs method. Equation (20.1.10) can be recast into a form which brings up the stabilization process of the unstable central scheme introduced by the upwinding process. Equation (20.1.10) is written as

$$u_i^{n+1} - u_i^n = -\frac{\sigma}{2}(u_{i+1}^n - u_{i-1}^n) + \frac{\tau}{2}|a|(u_{i+1}^n - 2u_i^n + u_{i-1}^n) \quad (20.1.14)$$

showing the presence of a numerical viscosity of the form $\Delta x^2 |\sigma| u_{xx}/2$ added to an otherwise centrally discretized scheme. Hence, two options can be taken in order to take properly into account the propagation properties of an hyperbolic equation. Either one applies an upwind, directionally biased space discretization, or one uses central discretization without paying attention to the direction of propagation, but introduces an adapted artificial viscosity term. As described in Chapter 15, this technique has been largely applied for steady transonic potential flow computations. As an additional comparison, the Lax-Wendroff scheme has the same form as (20.1.14) but with $\sigma^2/2$ as the coefficient of the centrally discretized u_{xx} term. Rewriting equation (20.1.10) in still another form, namely

$$u_i^{n+1} = \tau a^+ u_{i-1}^n + (1 - \tau|a|)u_i^n + (-\tau a^-)u_{i+1}^n \quad (20.1.15)$$

it is seen that all weight coefficients of the right-hand side contributions to the solution at level $n+1$ are positive when the CFL condition (20.1.11) is satisfied. This is illustrated in Figure 20.1.3.

This property is necessary and sufficient for a scheme to be *monotone* as defined by Godunov (1959). This will be discussed more in detail in Chapter 21. Let us, however, mention here that a monotone scheme has the property of not allowing the creation of new extrema and does not allow unphysical discontinuities. However, they cannot have an order of accuracy higher than one.

When applied to *linear* systems of equations, the scalar form of the first-order upwind scheme can be applied to each of the decoupled characteristic equations separately. With the diagonal matrix of the eigenvalues Λ , defined by equation (16.4.16), the schemes (20.1.14) written for the three characteristics can be

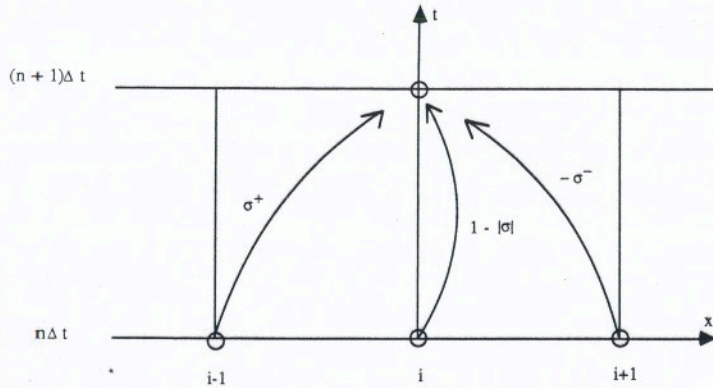


Figure 20.1.3 Weight coefficients in the distributive interpretation of the first-order upwind scheme

grouped as

$$W_i^{n+1} - W_i^n = -\frac{\tau}{2} \Lambda (W_{i+1}^n - W_{i-1}^n) + \frac{\tau}{2} |\Lambda| (W_{i+1}^n - 2W_i^n + W_{i-1}^n) \quad (20.1.16)$$

where W is the vector of the characteristic variables (16.4.18) and $|\Lambda|$ is the diagonal matrix of the absolute value of the eigenvalues. Reverting back to the conservative variables through the transformation (16.3.39) and defining the absolute value of the Jacobian by

$$|A| = P|\Lambda|P^{-1} \quad (20.1.17)$$

leads to the conservative form of the first-order upwind scheme

$$U_i^{n+1} - U_i^n = -\frac{\tau}{2} (f_{i+1}^n - f_{i-1}^n) + \frac{\tau}{2} |A| (U_{i+1}^n - 2U_i^n + U_{i-1}^n) \quad (20.1.18)$$

Note that scheme (20.1.18) has the numerical flux

$$f_{i+1/2}^* = \frac{f_{i+1} + f_i}{2} - |A| \frac{U_{i+1} - U_i}{2} \quad (20.1.19)$$

The stability condition is

$$\frac{\Delta t}{\Delta x} |\lambda_{\max}| \leq 1 \quad (20.1.20)$$

This scheme is uniquely defined by equation (20.1.18) when the system is linear. When applied to non-linear systems, it is not clear at which point the Jacobian matrix A has to be evaluated. In addition, one requires the schemes to be in conservative form. The flux splitting method of the following section is one way of combining upwind discretization with conservation.

20.2 ONE-DIMENSIONAL FLUX VECTOR SPLITTING

Since the Euler equations form a coupled system in the conservative variables and since the correct capture of discontinuities requires a conservative formulation, a more general definition of upwind schemes is necessary in order to define a splitting of the fluxes according to the signs of the eigenvalues. This can be achieved in the following way, introduced by Steger and Warming (1981).

20.2.1 Steger and Warming flux vector splitting

If the homogeneous Euler equations are put in characteristic form, equation (16.4.19),

$$\frac{\partial W}{\partial t} + \Lambda \frac{\partial W}{\partial x} = 0 \quad (20.2.1)$$

the upwind scheme (20.1.10) can be applied to each of the three characteristic variables separately, with the definitions

$$\lambda_k^+ = \frac{\lambda_k + |\lambda_k|}{2} \quad (20.2.2a)$$

$$\lambda_k^- = \frac{\lambda_k - |\lambda_k|}{2} \quad (20.2.2b)$$

for each of the eigenvalues of Λ :

$$\Lambda = \begin{vmatrix} \lambda_1 & \cdot & \cdot \\ \cdot & \lambda_2 & \cdot \\ \cdot & \cdot & \lambda_3 \end{vmatrix} = \begin{vmatrix} u & & \\ & u+c & \\ & & u-c \end{vmatrix} \quad (20.2.3)$$

This defines two diagonal matrices $\Lambda^\pm$:

$$\Lambda^\pm = \begin{vmatrix} \lambda_1^\pm & & \\ & \lambda_2^\pm & \\ & & \lambda_3^\pm \end{vmatrix} = \begin{vmatrix} \frac{u \pm |u|}{2} & & \\ & \frac{(u+c) \pm |u+c|}{2} & \\ & & \frac{(u-c) \pm |u-c|}{2} \end{vmatrix} \quad (20.2.4)$$

where Λ^+ has only positive eigenvalues, Λ^- only negative eigenvalues, and such that

$$\Lambda = \Lambda^+ + \Lambda^- \quad |\Lambda| = \Lambda^+ - \Lambda^- \quad (20.2.5)$$

or

$$\lambda_k = \lambda_k^+ + \lambda_k^- \quad |\lambda_k| = \lambda_k^+ - \lambda_k^- \quad (20.2.6)$$

The quasi-linear coupled equations are obtained from the characteristic form by the transformation matrix P , with the Jacobian A satisfying

$$A = P \Lambda P^{-1} \quad (20.2.7)$$

Hence an upwind formulation can be obtained with the Jacobians

$$A^+ = P \Lambda^+ P^{-1} \quad A^- = P \Lambda^- P^{-1} \quad (20.2.8)$$

with

$$A = A^+ + A^- \quad (20.2.9a)$$

$$|A| = A^+ - A^- \quad (20.2.9b)$$

The fluxes associated with these split Jacobians are obtained from the remarkable property of homogeneity of the flux vector $f(U)$. As noted earlier (Section 16.2), $f(U)$ is an homogeneous function of degree one of U . Hence, one has

$$f = A \cdot U \quad (20.2.10)$$

and the following flux splitting can be defined:

$$f^+ = A^+ \cdot U \quad f^- = A^- \cdot U \quad (20.2.11)$$

with

$$f = f^+ + f^- \quad (20.2.12)$$

This *flux vector* splitting, based on (20.2.2), has been introduced by Steger and Warming (1981). The split fluxes f^+ and f^- are also homogeneous functions of degree one in U .

It is important to notice that the splitting (20.2.2) is not unique, even under the restriction (20.2.5). For instance, the following splitting has been applied by Steger (1978):

$$\begin{aligned} \lambda_1^+ &= \frac{u + |u|}{2} & \lambda_1^- &= \frac{u - |u|}{2} \\ \lambda_2^+ &= \lambda_1^+ + c & \lambda_2^- &= \lambda_1^- \\ \lambda_3^+ &= \lambda_1^+ & \lambda_3^- &= \lambda_1^- - c \end{aligned} \quad (20.2.13)$$

which verifies equation (20.2.5), $\lambda_k^+ + \lambda_k^- = \lambda_k$.

Subsonic flow

For a subsonic flow, the third eigenvalue $\lambda_3 = u - c$ is negative and the two others are positive. Hence

$$\Lambda^+ = \begin{vmatrix} u & & \\ & u+c & \\ & & 0 \end{vmatrix} \quad \Lambda^- = \begin{vmatrix} 0 & & \\ & 0 & \\ & & u-c \end{vmatrix} \quad (20.2.14)$$

The Jacobians A^+ and A^- are obtained from (20.2.8) with the matrices P and

P^{-1} given by equations (16.4.11) and (16.4.12). Starting with A^- , which has more zero elements, one obtains (see Problem 20.1)

$$f^- = \frac{\rho(u-c)}{2\gamma} \begin{vmatrix} 1 \\ (u-c) \\ \frac{(u-c)^2}{2} + \left(\frac{3-\gamma}{\gamma-1}\right)\frac{c^2}{2} \end{vmatrix} \quad (20.2.15)$$

and f^+ is obtained from $f^+ = f - f^-$, where the energy has been expressed as a function of the sonic velocity by

$$\varepsilon = \rho E = \frac{\rho c^2}{\gamma(\gamma-1)} + \frac{\rho u^2}{2} \quad (20.2.16)$$

$$f^+ = \frac{\rho}{2\gamma} \begin{vmatrix} (2\gamma-1)u+c \\ 2(\gamma-1)u^2 + (u+c)^2 \\ (\gamma-1)u^3 + \frac{(u+c)^3}{2} + \frac{3-\gamma}{2(\gamma-1)}(u+c)c^2 \end{vmatrix} \quad (20.2.17)$$

A general calculation for an arbitrary splitting:

$$\bar{\Lambda} = \begin{vmatrix} \bar{\lambda}_1 & & \\ & \bar{\lambda}_2 & \\ & & \bar{\lambda}_3 \end{vmatrix} \quad (20.2.18)$$

performed by Steger and Warming, gives the corresponding flux as (see also Problem 20.2)

$$\bar{f} = P \bar{\Lambda} P^{-1} U = \frac{\rho}{2\gamma} \begin{vmatrix} 2(\gamma-1)\bar{\lambda}_1 + \bar{\lambda}_2 + \bar{\lambda}_3 \\ 2(\gamma-1)\bar{\lambda}_1 u + \bar{\lambda}_2(u+c) + \bar{\lambda}_3(u-c) \\ (\gamma-1)\bar{\lambda}_1 u^2 + \frac{\bar{\lambda}_2}{2}(u+c)^2 + \frac{\bar{\lambda}_3}{2}(u-c)^2 + \frac{3-\gamma}{2(\gamma-1)}(\bar{\lambda}_2 + \bar{\lambda}_3)c^2 \end{vmatrix} \quad (20.2.19)$$

Before presenting some schemes based on the flux vector splitting, it is important to notice a certain number of properties of the split fluxes.

20.2.2 Properties of split flux vectors

A first important remark to be made here is that the whole concept of flux splitting as defined by equations (20.2.9) to (20.2.11) is totally dependent on the fact that the fluxes are homogeneous functions of degree one in U . One cannot therefore directly apply this approach to a general scalar flux function $f(u)$, since the only homogeneous scalar flux function of degree one is the linear case $f = au$. For instance, the flux of Burgers equation $f = u^2/2$ is homogeneous of

degree two and has the Jacobian $a(u) = df/du = u$, leading to the relation $f = au/2$ instead of the scalar version of equation (20.2.10). Upwind methods that are not dependent on the homogeneity property of the fluxes will be discussed in Section 20.5 in relation to Godunov-type methods.

The splittings (20.2.9) and (20.2.11) give

$$\begin{aligned} A &= A^+ + A^- \\ f &= AU = f^+ + f^- = A^+ U + A^- U \end{aligned} \quad (20.2.20)$$

and

$$\frac{\partial f}{\partial U} = A = \frac{\partial f^+}{\partial U} + \frac{\partial f^-}{\partial U} = A^+ + A^- \quad (20.2.21)$$

but one has *not* generally the equality of the split Jacobians, that is

$$\frac{\partial f^+}{\partial U} \neq A^+ \quad \text{and} \quad \frac{\partial f^-}{\partial U} \neq A^- \quad (20.2.22)$$

This can be verified by a direct calculation for the Steger–Warming flux splitting (see Problem 20.4).

More importantly, these matrices have not the same set of eigenvalues. By construction, the eigenvalues of A^+ and A^- are the positive and negative eigenvalues of A , that is λ_k^+ and λ_k^- , but this is not true for $\partial f^+/\partial U$ and $\partial f^-/\partial U$.

However, Steger and Warming (1981) report that $f_u^+ \equiv \partial f^+/\partial U$ has only positive eigenvalues and $f_u^- \equiv \partial f^-/\partial U$ has only negative eigenvalues. Lerat (1983) has given an analytical proof of this property for the one-dimensional Euler equations for a perfect gas, under the condition that the specific heat ratio γ satisfies the condition $1 < \gamma < \frac{5}{3}$ (see also Problem 20.5).

The non-equality of f_u^+ and A^+ on the one hand and of f_u^- and A^- on the other hand has very important consequences with regard to the definition of upwind schemes based on the concept of flux vector splitting. Indeed, we can write the conservation equations in split form as

$$\frac{\partial U}{\partial t} + \frac{\partial f^+}{\partial x} + \frac{\partial f^-}{\partial x} = 0 \quad (20.2.23)$$

which becomes, in quasi-linear form,

$$\frac{\partial U}{\partial t} + f_u^+ \frac{\partial U}{\partial x} + f_u^- \frac{\partial U}{\partial x} = 0 \quad (20.2.24)$$

Alternatively, we can first write the quasi-linear form and split the Jacobians afterwards, that is

$$\frac{\partial U}{\partial t} + A \frac{\partial U}{\partial x} = \frac{\partial U}{\partial t} + A^+ \frac{\partial U}{\partial x} + A^- \frac{\partial U}{\partial x} = 0 \quad (20.2.25)$$

These two formulations are not identical, as a consequence of equations (20.2.22), showing that the two operations of splitting the fluxes and introducing the

quasi-linearization do not commute. Only (20.2.25) corresponds to the direct generalization of the uncoupled equations (20.1.10) and therefore keeps the same physical characteristics which are separated according to their sign. Equation (20.2.24) does not represent a decomposition of the physical characteristics, but is the only linearization consistent with the conservative form (20.2.23).

Connection with the eigenvectors of the Jacobian matrix A

The split fluxes can be related to the eigenvectors of the Jacobian matrix A , in particular to the right eigenvectors, that constitute the columns of the matrix P (see Section 16.4). This leads to a simple way of computing the Jacobian $A^\pm$ and the associated fluxes $f^\pm$ (Lerat, 1983).

The set of left (or right) eigenvectors $l^{(j)}$ (or $r^{(j)}$) associated with the eigenvalue λ_j of A are linearly independent by definition of the hyperbolic character of the Euler equations. Hence, any function of the conservative variables can be expressed as linear combinations of the left (or the right) eigenvectors considered as basic vectors spanning the U space. In particular, we have

$$U = \sum_{j=1}^3 \alpha_j(U) r^{(j)}(U) \quad (20.2.26)$$

where α_j are the coefficients of the expansion and can be considered as the projection of U in the base $r^{(j)}$.

Since the $r^{(j)}$ vectors are defined up to a normalization constant (see equation (16.4.8)), we can always normalize each $r^{(j)}$ such that $\alpha_j = 1$. The appropriate normalization factors can be derived by inspection of equation (16.4.12) which defines the matrix P and which is repeated here for convenience. The matrix P

$$P = \begin{vmatrix} 1 & \frac{\rho}{2c} & -\frac{\rho}{2c} \\ u & \frac{\rho(u+c)}{2c} & -\frac{\rho(u-c)}{2c} \\ \frac{u^2}{2} & \frac{\rho(H+uc)}{2c} & -\frac{\rho(H-uc)}{2c} \end{vmatrix} \quad (20.2.27)$$

contains the right eigenvectors $r^{(j)}$ as columns with the normalization defined by $\alpha = \beta = \delta = 1$ in equation (16.4.8).

If $\alpha_j = 1$, equation (20.2.26) shows that the sum of the elements along the same line of P must be equal to the elements of U . This leads to the following renormalized eigenvectors $r^{(j)}$ (see Problem 16.24):

$$r^{(1)} = \rho \frac{\gamma-1}{\gamma} \begin{vmatrix} 1 \\ u \\ u^2 \\ 2 \end{vmatrix} \quad r^{(2)} = \frac{\rho}{2\gamma} \begin{vmatrix} 1 \\ u+c \\ H+uc \end{vmatrix} \quad r^{(3)} = \frac{\rho}{2\gamma} \begin{vmatrix} 1 \\ u-c \\ H-uc \end{vmatrix} \quad (20.2.28)$$

where the perfect gas relations

$$p = (\gamma - 1)\rho e$$

$$E = e + \frac{u^2}{2} = H - \frac{p}{\rho} \quad (20.2.29)$$

have been introduced. Equation (20.2.26) now becomes

$$U = \sum_{j=1}^3 r^{(j)} \quad (20.2.30)$$

The flux vector f can also be projected in the space $r^{(j)}$, applying the homogeneity property of the fluxes in the Euler equations,

$$f = AU = \sum_{j=1}^3 Ar^{(j)} \quad (20.2.31)$$

Since r^j is an eigenvector of A associated with the eigenvalue λ_j ,

$$f = \sum_{j=1}^3 \lambda_j r^{(j)} \quad (20.2.32)$$

showing that the projections of the flux vector on the basis $r^{(j)}$ are precisely the corresponding eigenvalues. This can be verified by a direct calculation (see Problem 20.7).

The split fluxes can now be defined with equations (20.2.11) and (20.2.2) to (20.2.9) as follows:

$$f = \sum_{j=1}^3 (\lambda_j^+ + \lambda_j^-) r^{(j)} = f^+ + f^- \quad (20.2.33)$$

with

$$f^+ = \sum_{j=1}^3 \lambda_j^+ r^{(j)} \quad f^- = \sum_{j=1}^3 \lambda_j^- r^{(j)} \quad (20.2.34)$$

This applies also to the arbitrary splitting of equation (20.2.18), where the flux $\bar{f}$, associated with a set of partial eigenvalues $\bar{\lambda}_j$, becomes

$$\bar{f} = \sum_{j=1}^3 \bar{\lambda}_j r^{(j)} \quad (20.2.35)$$

which should be identical to equation (20.2.19).

20.2.3 Van Leer's flux splitting

The Jacobian of the split fluxes f_u^+ and f_u^- as defined above are not continuously differentiable, since they have a discontinuous slope at sonic velocities. This can be best be seen from the mass flux components, expressed as a function of the variables $\rho, c, M = u/c$, introducing the Mach number M . Since $f_1 = \rho u = \rho c M$ is linear in M , a plot of $f_1/\rho c$ as a function of Mach number illustrates the properties we wish to put forward. Since $f^+ = f$ for supersonic flows $u > c$, and

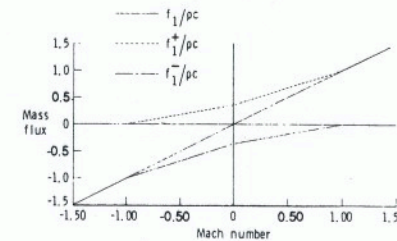
considering negative as well as positive velocities, we have (see also Problem 20.7)

$$f_1^+ = \begin{cases} 0 & \text{for } M \leq -1 \\ \frac{\rho c}{2\gamma}(M+1) & \text{for } -1 \leq M \leq 0 \\ \frac{\rho c}{2\gamma}[(2\gamma-1)M+1] & \text{for } 0 \leq M \leq 1 \\ f_1 & \text{for } M > 1 \end{cases} \quad (20.2.36)$$

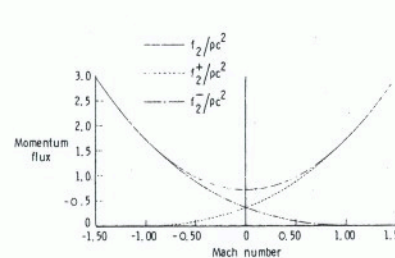
Similarly,

$$f_1^- = \begin{cases} f_1 & \text{for } M \leq -1 \\ \frac{\rho c}{2\gamma}[(2\gamma-1)M-1] & \text{for } -1 \leq M \leq 0 \\ \frac{\rho c}{2\gamma}(M-1) & \text{for } 0 \leq M \leq 1 \\ 0 & \text{for } M > 1 \end{cases} \quad (20.2.37)$$

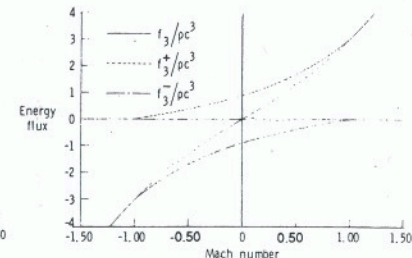
The variations of $f_1^\pm$ are shown in Figure 20.2.1(a). The discontinuity in the



(a) Mass flux



(b) Momentum flux



(c) Energy flux

Figure 20.2.1 Mach number dependence of the Steger-Warming split fluxes. (From Anderson et al., 1986a)

slope of $f_1^\pm$ when the eigenvalues go to zero (at sonic and stagnation points) is clearly seen. Figures 20.2.1(b) and 20.2.1(c) show similarly the momentum fluxes $f_2^\pm$ divided by ρc^2 and the energy fluxes divided by ρc^3 , both as a function of Mach number.

This will cause certain difficulties since a discontinuity in slope at the sonic transition might occur on the computed solution (Steger and Warming, 1981). This is illustrated later in Figure 20.3.7, which shows the results of a shock tube flow computation with a first-order upwind scheme based on the Steger-Warming flux splitting. The 'glitch' at the sonic transition is clearly seen.

Van Leer (1982) has introduced a flux splitting different from (20.2.15), (20.2.17) by imposing a certain number of conditions on f^+ and f^- satisfying (20.2.12). In particular, $f^\pm$ and the associated Jacobians $f_u^\pm$ are requested to be continuous functions of Mach number and expressed as polynomials of the lowest possible order. In addition, the eigenvalues of f_u^+ must be positive or zero and those of f_u^- negative or zero, with one eigenvalue equal to zero in the subsonic range $|M| < 1$. This last condition is fulfilled by the Jacobians A^+ and A^- but generally not by f_u^+ and f_u^- .

Since for $M \geq 1$, $f^+ = f$ and for $M \leq -1$, $f^- = f$, the symmetry properties of each flux component $f_j^\pm$ must be the same as those of the total flux, that is

$$f_j^+(M) = \pm f_j^-(-M) \quad \text{if } f_j(M) = \pm f_j(-M) \quad (20.2.38)$$

The above conditions require the flux components to be proportional to $(M \pm 1)^n$, with $n \geq 2$ for vanishing slopes at $M = \mp 1$. The lowest possible order is $n = 2$. The simplest expression for the mass flux component is then $f_1^\pm(M) = \rho c(M \pm 1)^2/4$. The momentum flux component can be defined as $f_2^\pm = f_1^\pm \cdot P(M)$, where $P(M)$ is the lowest possible polynomial in M , namely first order. The energy flux component is selected under the condition that one of the eigenvalues of the split flux Jacobians $f_u^\pm$ should vanish in the range $0 < M < 1$. Hence $f_3^\pm$ is taken to be proportional to $f_2^{\pm 2}/f_1^\pm$. This leads to the following form (Van Leer, 1982):

$$f_{\text{VL}}^+ = \frac{\rho}{4c}(u+c)^2 \begin{vmatrix} 1 \\ \frac{(\gamma-1)u+2c}{\gamma} \\ \frac{[(\gamma-1)u+2c]^2}{2(\gamma^2-1)} \end{vmatrix} \\ = \frac{\rho c}{4}(M+1)^2 \begin{vmatrix} 1 \\ \frac{2c}{\gamma} \left(1 + \frac{\gamma-1}{2}M\right) \\ \frac{2c^2}{\gamma^2-1} \left(1 + \frac{\gamma-1}{2}M\right)^2 \end{vmatrix} \quad (20.2.39)$$

The negative part of the flux is given by

$$f_{\text{VL}}^- = f - f_{\text{VL}}^+ = -\frac{\rho c}{4}(M-1)^2 \begin{vmatrix} 1 \\ \frac{2c}{\gamma} \left(-1 + \frac{\gamma-1}{2}M\right) \\ \frac{2c^2}{\gamma^2-1} \left(1 - \frac{\gamma-1}{2}M\right)^2 \end{vmatrix} \\ = -\frac{\rho}{4c}(u-c)^2 \begin{vmatrix} 1 \\ \frac{(\gamma-1)u-2c}{\gamma} \\ \frac{[2c-(\gamma-1)u]^2}{2(\gamma^2-1)} \end{vmatrix} \quad (20.2.40)$$

One has clearly $f_{j,\text{VL}}^+(M) = -f_{j,\text{VL}}^-(-M)$ for the first ($j=1$) and third ($j=3$) components, which are uneven in M ($f_1 = \rho u = \rho c M$ and $f_3 = \rho u H = \rho c^3 M[1/(\gamma-1) + M^2/2]$) and $f_{2,\text{VL}}^+(M) = f_{2,\text{VL}}^-(-M)$ for the momentum flux component $f_2 = \rho u^2 + p = \rho c^2(1 + \gamma M^2)/\gamma$, which is an even function of M . The continuity of the slopes $\partial f^+/\partial M$ at $M=1$ is readily checked, while at $M=-1$ it is guaranteed by the presence of the factor $(M+1)^2$, which has a vanishing derivative at $M=-1$ (see also Problem 20.9).

The Van Leer split flux components are represented in Figure 20.2.2 as a function of M , and the continuity of slopes at sonic and stagnation points is clearly seen.

A direct but lengthy calculation, identifying the obtained splitting with the general form (20.2.19), leads to the following eigenvalue decomposition:

$$\lambda_{1,\text{VL}}^+ = \frac{c}{4}(M+1)^2 \left[1 - \frac{(M-1)^2}{\gamma+1} \right] \quad (20.2.41a)$$

$$\lambda_{2,\text{VL}}^+ = \frac{c}{4}(M+1)^2 \left[3 - M + \frac{\gamma-1}{\gamma+1}(M-1)^2 \right] \quad (20.2.41b)$$

$$\lambda_{3,\text{VL}}^+ = \frac{c}{2(\gamma+1)}(M+1)^2(M-1) \left(1 + \frac{\gamma-1}{2}M \right) \quad (20.2.41c)$$

Note that $\lambda_{3,\text{VL}}^+$ is negative in the range $0 < M < 1$, although the eigenvalues of the Jacobian of the corresponding split flux are positive or zero.

The negative parts are obtained from

$$\begin{aligned} \lambda_{1,\text{VL}}^- &= -\lambda_{1,\text{VL}}^+(-M) \\ \lambda_{2,\text{VL}}^- &= -\lambda_{2,\text{VL}}^+(-M) \\ \lambda_{3,\text{VL}}^- &= -\lambda_{3,\text{VL}}^+(-M) \end{aligned} \quad (20.2.42)$$

A direct calculation shows that the relations (20.2.6) are satisfied.

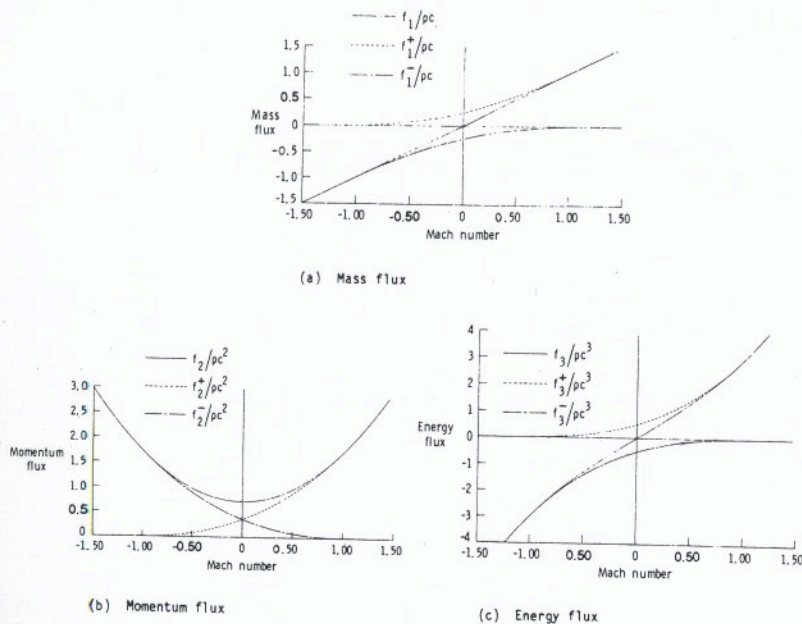


Figure 20.2.2 Mach number dependence of the Van Leer split fluxes. (From Anderson *et al.*, 1986a)

This eigenvalue splitting is obviously strongly distinct from the Steger–Warming choice. The vanishing of one of the eigenvalues of $f_{u,VL}^{\pm}$ for $|M| \leq 1$ forces a steady shock transition over two mesh cells (Van Leer, 1982).

A variant of the splitting (20.2.39), (20.2.40) has been applied by Hänel *et al.* (1987), with the aim of ensuring the constancy of the stagnation enthalpy H for stationary flows. Referring to Figure 20.3.1, the steady-state solution of a first-order upwind discretization is obtained from the balance of fluxes $f_{i-1}^+ + f_i^- = f_i^+ + f_{i+1}^-$ and when applied to the energy component the above splittings do not lead to the constancy of H over the cell $i(i-1/2, i+1/2)$ (see Problem 20.14). This is a weakness of the flux splitting approach, which is not shared by the central schemes which solve $f_{i-1} = f_{i+1}$, leading to $H_{i-1} = H_{i+1}$ when mass conservation is satisfied. The error is, however, of the order of the truncation error.

Since the energy flux $f_3 = \rho u H = \rho c M H$, a valid splitting of the third component is

$$f_3^{\pm} = f_1^{\pm} H \quad (20.2.43)$$

The third component of (20.2.39) can be written as $f_3^{\pm} = f_1^{\pm} [H - (M \mp 1)^2 c^2 / (\gamma + 1)]$, indicating that the choice (20.2.43) satisfies the conditions of slope

continuity at sonic and stagnation points. However, there is no longer any vanishing eigenvalue of the split flux Jacobians in the range $|M| \leq 1$; refer also to Problem 20.10.

The energy flux in the Van Leer splitting can be generalized to the form

$$f_3^{\pm} = f_1^{\pm} [H - c^2 P(M \pm 1)] \quad (20.2.44)$$

where $P(M \pm 1)$ is a polynomial in $M \pm 1$. If it is requested that the slopes $\partial f^{\pm} / \partial M$ are continuous at $M = 1$, then P should be a polynomial in $(M - 1)$, for f^+ of second degree or higher. The lowest degree is two, leading to the original choice of (20.2.39). Other properties could be put forward (see, for instance, Problem 20.15), and use can be made of this degree of freedom for extending the flux splittings to non-perfect gases (Vinokur and Liu, 1988; Liou *et al.*, 1988).

20.2.4 Non-reflective boundary conditions and split fluxes

The non-equality of $A^{\pm}$ and $f_u^{\pm}$ has also an impact on the treatment of boundary conditions by methods connected to the characteristic approach. The compatibility relations for the outgoing characteristics are written as follows, referring to Section 19.1,

$$\frac{\partial W^N}{\partial t} + \Lambda \frac{\partial W^N}{\partial x} = P^{-1} Q \quad (20.2.45)$$

where the superscript N stands for numerical boundary condition and where W^N represents the number of outgoing characteristics.

At a subsonic inlet, W^N stands for the third characteristic w_3 and (20.2.45) represents one equation, while at a subsonic exit W^N stands for the characteristics w_1 and w_2 so that (20.2.45) represents two equations. To fix our ideas, let us suppose that the outgoing characteristics correspond to positive eigenvalues. The above equation can then be written as

$$\frac{\partial W^N}{\partial t} + \Lambda^+ \frac{\partial W^N}{\partial x} = P^{-1} Q \quad (20.2.46)$$

The non-reflecting condition (19.1.78) for the incoming characteristics is then associated with the negative eigenvalues and corresponds to the condition $\partial W^P / \partial t = P^{-1} Q$. Hence, when combined with the above compatibility equation as described at the end of Section 19.1.5, the three equations obtained in this way can be written as

$$\frac{\partial W}{\partial t} + \Lambda^+ \frac{\partial W}{\partial x} = P^{-1} Q \quad (20.2.47)$$

Reverting back to conservative variables, one obtains, in the presence of a source term Q , where A^+ is defined from the Steger–Warming flux splitting

(20.2.2), the non-conservative formulation

$$\frac{\partial U}{\partial t} + A^+ \frac{\partial U}{\partial x} = Q \quad (20.2.48)$$

These boundary conditions have to be discretized in an upwind way, for instance at an exit boundary point $i = M$,

$$U_M^{n+1} - U_M^n = -\tau A_M^+(U_M - U_{M-1}) + \Delta t \cdot Q_M \quad (20.2.49)$$

On the other hand, the alternative conservative formulation of the numerical conditions

$$\frac{\partial U}{\partial t} + \frac{\partial f^+}{\partial x} = Q \quad (20.2.50)$$

discretized as

$$U_M^{n+1} - U_M^n = -\tau(f_M^+ - f_{M-1}^+) + \Delta t \cdot Q \quad (20.2.51)$$

has not the same physical significance, since the eigenvalues of f_u^+ are not equal to the eigenvalues Λ^+ of A^+ . Hence along the path of the f_u^+ eigenvalues, the Riemann invariants are not constant as is the case when applying equation (20.2.48). Therefore, this formulation is not physically consistent and *should not be applied* with the split flux vectors f^+ and f^- .

When the incorrect boundary treatment (20.2.51) is applied, for instance to the one-dimensional shock tube problem, reflected waves are produced at the exit boundary. This will not be the case with the formulation (20.2.49). Therefore, equation (20.2.48) represents a consistent formulation for the compatibility relations in combination with the non-reflective condition.

20.3 ONE-DIMENSIONAL UPWIND DISCRETIZATIONS BASED ON FLUX VECTOR SPLITTING

The definition of fluxes with Jacobian eigenvalues always having a positive or negative sign, like f^+ and f^- , allows a general definition of upwind schemes in conservation form, although these eigenvalues are not equal to those of A .

Writing the system of Euler equations in the form of equation (20.2.23)

$$\frac{\partial U}{\partial t} + \frac{\partial f^+}{\partial x} + \frac{\partial f^-}{\partial x} = 0 \quad (20.3.1)$$

the basic idea of the upwind scheme will be realized if f_x^+ is discretized with a backward difference and f_x^- with a forward difference.

20.3.1 First-order explicit upwind schemes

The following explicit first-order scheme is the obvious conservative generalization of the upwind scheme (20.1.10) (Steger and Warming, 1981):

$$\begin{aligned} U_i^{n+1} - U_i^n &= -\tau(f_i^+ - f_{i-1}^+) - \tau(f_{i+1}^- - f_i^-) \\ &= -\tau[f_i^+ + f_{i+1}^- - (f_{i-1}^+ + f_i^-)] \end{aligned} \quad (20.3.2)$$

This scheme is first-order accurate in space and time. It can also be written, after isolating the central difference terms, like in (20.1.18) by introducing the absolute value flux

$$|f| = f^+ - f^- = |A|U \quad (20.3.3)$$

where

$$|A| = P|\Lambda|P^{-1} \quad (20.3.4)$$

$$U_i^{n+1} - U_i^n = -\frac{\tau}{2}(f_{i+1} - f_{i-1})^n + \frac{\tau}{2}(|f|_{i+1} - 2|f|_i + |f|_{i-1})^n \quad (20.3.5)$$

showing an artificial viscosity term proportional to the second derivative of the absolute flux value $|f|$.

Comparing with the Lax-Wendroff scheme, we can transform equation (20.3.2) in a form that explicits the absolute value Jacobian $|A|$. Up to second-order accuracy, one can write, after a Taylor expansion valid for smooth variations,

$$|f|_{i+1} - |f|_i = |A|_{i+1/2}(U_{i+1} - U_i) + O(\Delta x)(U_{i+1} - U_i) \quad (20.3.6)$$

Hence, equation (20.3.2) becomes, to the same first-order accuracy,

$$\begin{aligned} U_i^{n+1} - U_i^n &= -\frac{\tau}{2}(f_{i+1} - f_{i-1})^n + \frac{\tau}{2}[|A|_{i+1/2}(U_{i+1} - U_i) \\ &\quad - |A|_{i-1/2}(U_i - U_{i-1})]^n + O(\Delta U^2) \end{aligned} \quad (20.3.7)$$

while the Lax-Wendroff scheme, which is second-order accurate, can be written

$$U_i^{n+1} - U_i^n = -\frac{\tau}{2}(f_{i+1} - f_{i-1})^n + \frac{\tau^2}{2}[A_{i+1/2}^2(U_{i+1} - U_i) - A_{i-1/2}^2(U_i - U_{i-1})]^n \quad (20.3.8)$$

The differences between the two schemes can be viewed as the replacement of the first-order coefficients $\tau|A|/2$ by a second-order coefficients $\tau^2 A^2/2$ in the pseudo-viscosity terms, which stabilize the unstable central difference scheme (see also Problem 20.10).

Equations (20.3.2) and (20.3.7) are equivalent formulations of the first-order conservative upwind scheme for smooth variations of U . In terms of a numerical flux f^* , one can write the conservative upwind scheme (20.3.2) as

$$U_i^{n+1} - U_i^n = -\tau \delta f_{i+1/2}^* \quad (20.3.9)$$

with

$$f_{i+1/2}^* = f_i^+ + f_{i+1}^- \quad (20.3.10a)$$

or, up to second-order accuracy,

$$f_{i+1/2}^* = \frac{1}{2}(f_i + f_{i+1}) - \frac{1}{2}A_{i+1/2}(U_{i+1} - U_i) + O(\Delta U^2) \quad (20.3.10b)$$

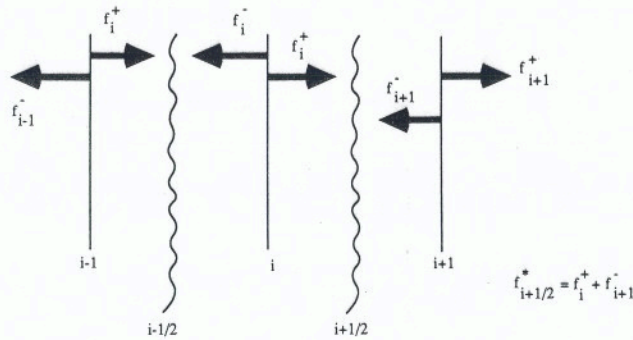


Figure 20.3.1 Interpretation of numerical fluxes associated with the first-order upwind scheme

This has an interesting physical interpretation, as seen from Figure 20.3.1. The flux contribution to the time variation of the solution at point i is described by the balance between the fluxes $f_{i+1/2}^*$ and $f_{i-1/2}^*$. If the fluxes f^+ and f^- are attached to the characteristics of positive and negative signs, it is seen that $f_{i+1/2}^*$ is the sum of the fluxes entering the cell $(i, i+1)$, namely f_i^+ and f_{i+1}^- , and similarly for $f_{i-1/2}^*$, which is formed by the fluxes entering the cell $(i, i-1)$.

20.3.2 Stability conditions for first-order flux vector splitting schemes

The stability conditions of scheme (20.3.2) have to be considered carefully. In the strict linear case, $f = au$, the stability condition reduces to the CFL condition of equation (20.1.11), $|\sigma| \leq 1$. However, in the non-linear case of equation (20.3.2), the Jacobians of $f^\pm$ are not equal to $A^\pm$, as seen in the previous section, and, even more significant, their eigenvalues are different. Hence, if a local linearization is performed, around a given state U^n , equation (20.3.2) becomes

$$U_i^{n+1} - U_i^n = -\tau f_u^{+n}(U_i - U_{i-1})^n - \tau f_u^{-n}(U_{i+1} - U_i)^n \quad (20.3.11)$$

where f_u^{+n} and f_u^{-n} are the linearized values taken by the Jacobians of f^+ and f^- for $U = U^n$. The amplification matrix G becomes, from a Von Neumann analysis,

$$G - 1 = -2\tau(f_u^+ - f_u^-) \sin^2 \frac{\phi}{2} - I\tau(f_u^+ + f_u^-) \sin \phi \quad (20.3.12)$$

Introducing the matrix

$$|f_u| = f_u^+ - f_u^- \quad (20.3.13)$$

which is different from $|A|$ because of equation (20.2.22), one has

$$G = 1 - 2\tau|f_u| \sin^2 \frac{\phi}{2} - I\tau A \sin \phi \quad (20.3.14)$$

The stability condition is given by the requirement of the spectral radius of G being lower or equal to one.

A difficulty arises here since $|f_u|$ and A have, generally, not the same set of eigenvalues. However, an estimation of the eigenvalues of the matrix G can be obtained by first transforming G with the similarity transformation matrix P which diagonalizes A . Since G and

$$\bar{G} = P^{-1}GP \quad (20.3.15)$$

have the same eigenvalues, one can as well analyse the eigenvalues of

$$\bar{G} = 1 - 2\tau|E| \sin^2 \frac{\phi}{2} - I\tau\Lambda \sin \phi \quad (20.3.16)$$

where $|E|$ is the transformed matrix of $|f_u|$ and Λ the diagonal matrix of the eigenvalues of A :

$$|E| = P^{-1}|f_u|P = P^{-1}f_u^+P - P^{-1}f_u^-P \quad (20.3.17)$$

A numerical estimation of the eigenvalues of $\bar{G}$ shows that they reach their maximum value for the high-frequency limit $\phi = \pi$. Hence, the maximum eigenvalue of $\bar{G}$, or its spectral radius $\rho(\bar{G})$, is given by

$$\rho(\bar{G}) = 1 - 2\tau\rho(|E|) \quad (20.3.18)$$

where $\rho(|E|)$ is the spectral radius of $|E|$. The stability condition of the flux split scheme (20.3.2) is therefore given by $\rho(\bar{G}) \leq 1$, that is by

$$\tau|e|_{\max} \leq 1 \quad (20.3.19)$$

where $|e|_{\max}$ is the maximum eigenvalue of $|E|$, instead of the linear CFL condition, written for $u > 0$:

$$\text{CFL} = \tau|u + c|_{\max} = \tau|\lambda|_{\max} \leq 1 \quad (20.3.20)$$

where $|\lambda|_{\max}$ represents the maximum eigenvalue of the Jacobian A . Hence, the stability limits of the scheme (20.3.2) are defined by the eigenvalues of $|f_u|$ and not by the Jacobian eigenvalues of $|A|$.

An explicit calculation performed by Lerat (1983) for a shock tube problem with an initial pressure ratio of 2.8 and equal temperatures on both sides of the diaphragm shows that the condition (20.3.19) corresponds to a CFL number limit $\tau|u + c|_{\max}$ below one; in particular, for this case the stability limit (20.3.19) corresponds to $\text{CFL} \leq 0.858$ instead of one according to equation (20.3.20).

Figure 20.3.2 presents computed results for this shock tube problem at $\text{CFL} = 0.9$ and 0.85 . The instability of the computed solutions at $\text{CFL} = 0.90$ clearly demonstrates the validity of the stability limit (20.3.19).

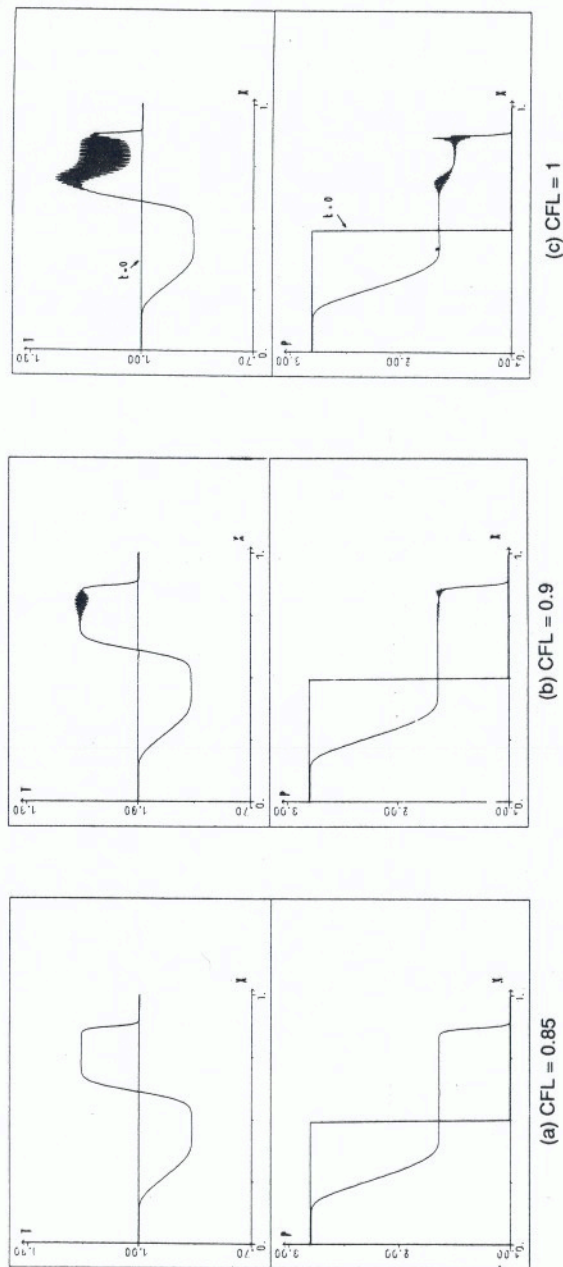


Figure 20.3.2 Shock tube problem with first-order Steger-Warming flux splitting. (From Lerat, 1983)

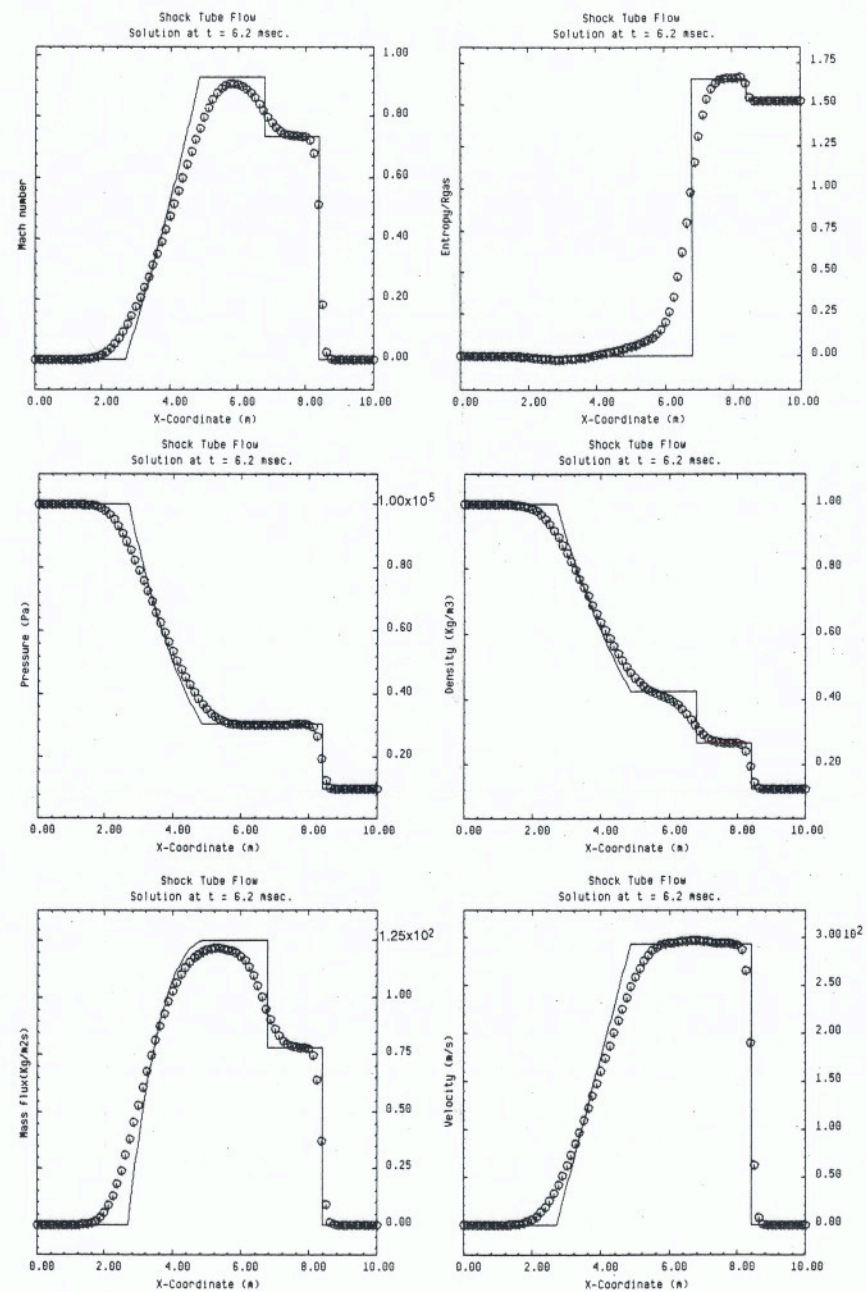


Figure 20.3.3 Computation of the shock tube problem with a first-order scheme and the flux vector splitting of Steger and Warming

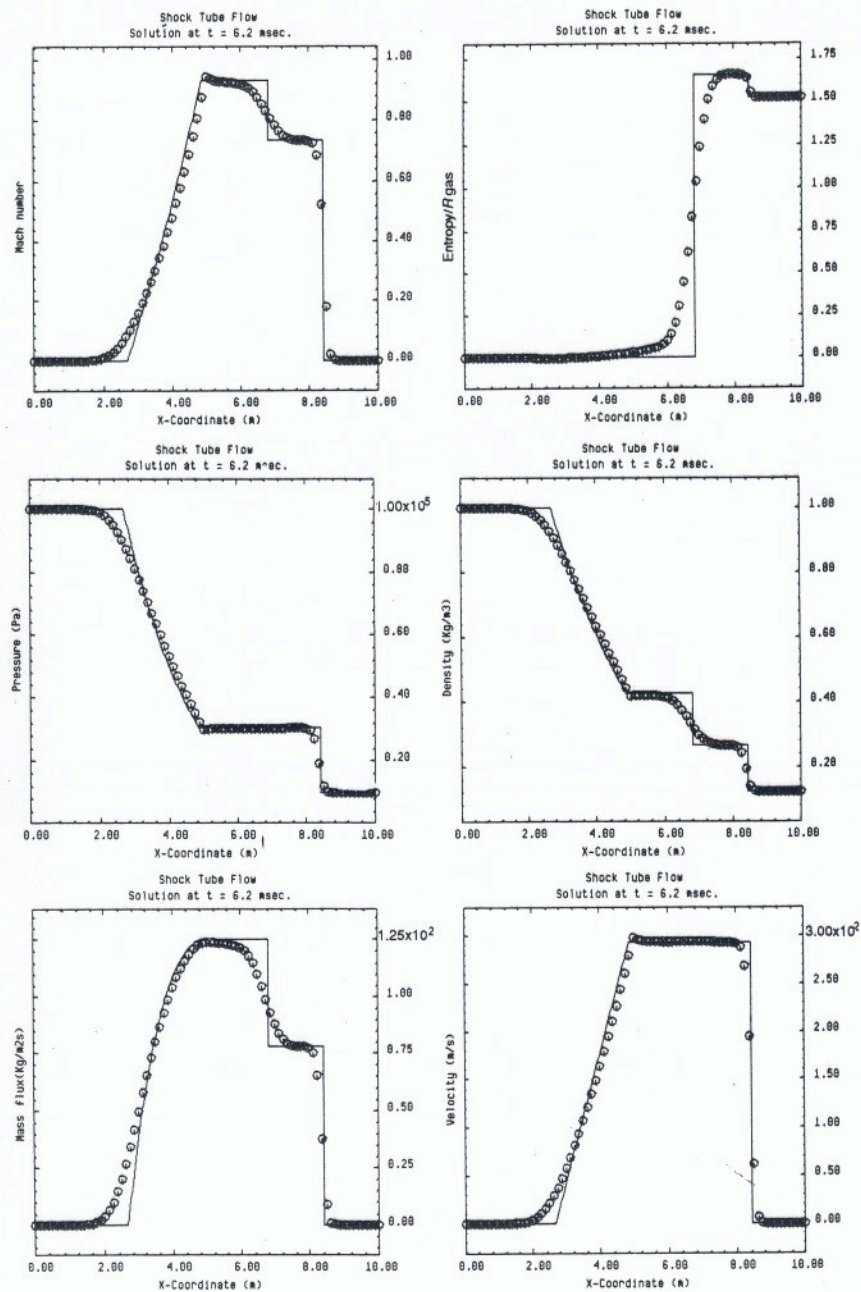


Figure 20.34 Computation of the shock tube problem with a first-order scheme and the flux vector splitting of Van Leer

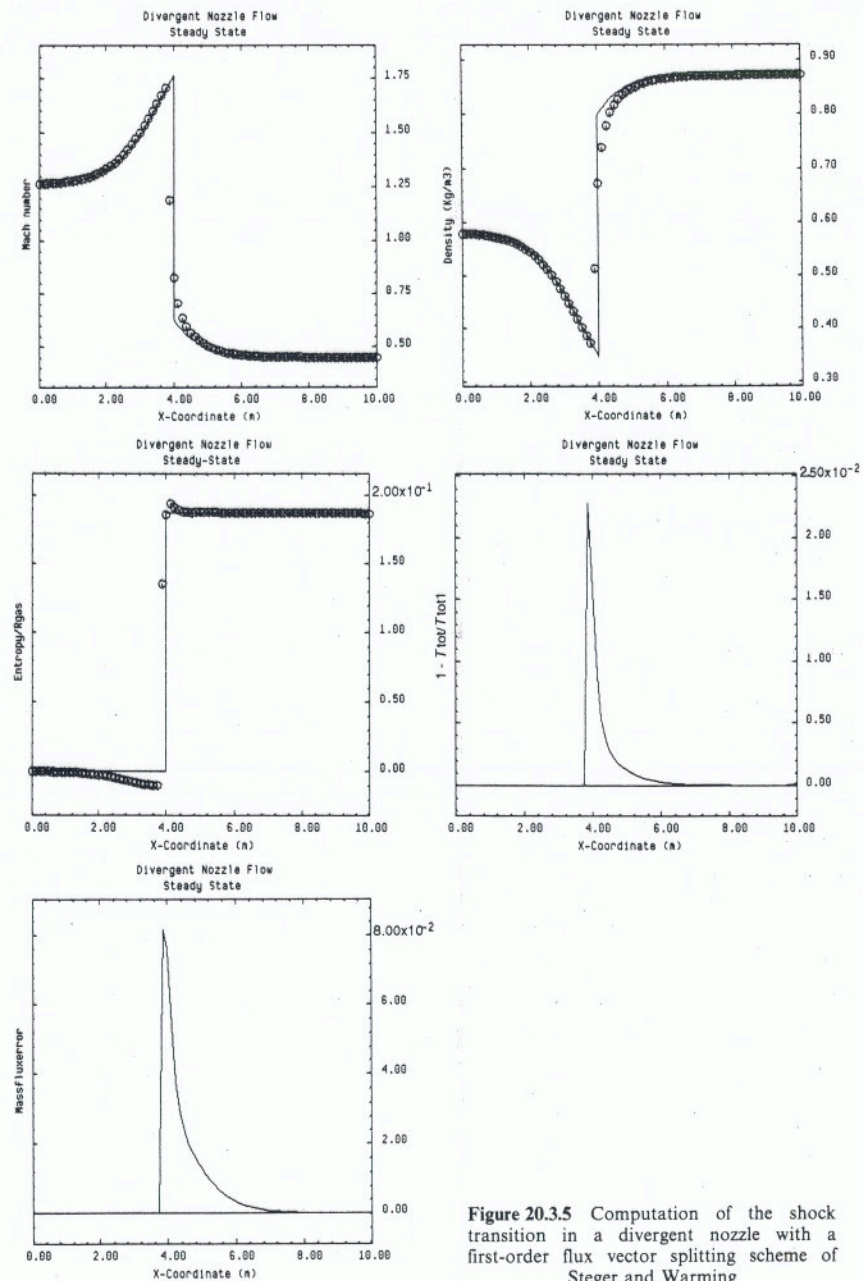


Figure 20.35 Computation of the shock transition in a divergent nozzle with a first-order flux vector splitting scheme of Steger and Warming

For the Van Leer splitting, a practical stability condition is given by Van Leer (1982), for $u > 0$:

$$\tau(u+c)_{\max} \leq \frac{2\gamma + M(3-\gamma)}{\gamma+3} \quad \text{for } M < 1 \quad (20.3.21)$$

Figures 20.3.3 and 20.3.4 show a comparison, on the shock tube problem with a pressure ratio of 10 after 35 time steps, between the Steger and Warming flux vector splitting and the flux splitting defined by Van Leer, both at $CFL = 0.95$ with the first-order upwind scheme (20.3.2). Van Leer's fluxes give a marked improvement of the expansion region, although in both cases the contact discontinuity is poorly represented.

This is a general property of flux splitting schemes, since sharp transitions require a vanishing dissipation when crossing discontinuities. Since this does not happen at contact discontinuities, they will tend to diffuse continuously with time. To circumvent this difficulty non-linear devices will have to be introduced (see Chapter 21). Note also that Roe's and Osher's approximate Riemann solvers, to be presented in Section 20.5, detect stationary contact discontinuities and hence better preserve their sharpness.

With regard to shocks, the Van Leer splitting has, by construction, a vanishing eigenvalue of the split flux Jacobians in the range $0 < M < 1$ and leads to sharp stationary shock transitions, over two cells. Since the Steger–Warming splitting does not have this property, shocks will be more smeared out. These properties are clearly illustrated in Figures 20.3.5 and 20.3.6, comparing the two methods on a shock transition in a divergent nozzle. Observe that in both cases there is no mass flux, nor total temperature error upstream of the shock. This results from the adaptation of the upwind procedure with the supersonic flow in this region, preventing errors to propagate upstream.

Figure 20.3.7 shows the results of a computation with the Steger–Warming flux splitting for a shock tube problem with an initial pressure discontinuity of 100, corresponding to the test case of Figure 16.6.9. The expansion fan reaches supersonic velocities and a sonic transition occurs at the initial position of the diaphragm $x = 5$.

Observe the small discontinuity at the sonic point transition as a consequence of the slope discontinuity of the components f^+ and f^- . This can be corrected by a modification proposed by Buning and Steger (1982), whereby the eigenvalues are redefined as

$$\lambda^\pm = \frac{\lambda \pm \sqrt{\lambda^2 + \varepsilon^2}}{2} \quad (20.3.22)$$

where ε is a small number. This leads to a smooth transition through the sonic points, as can be seen from Figure 20.3.8, at the expense of introducing some small error.

First-order schemes are not sufficiently accurate for practical purposes and the passage to second-order schemes is essential. This will be discussed in the following Chapter 21.

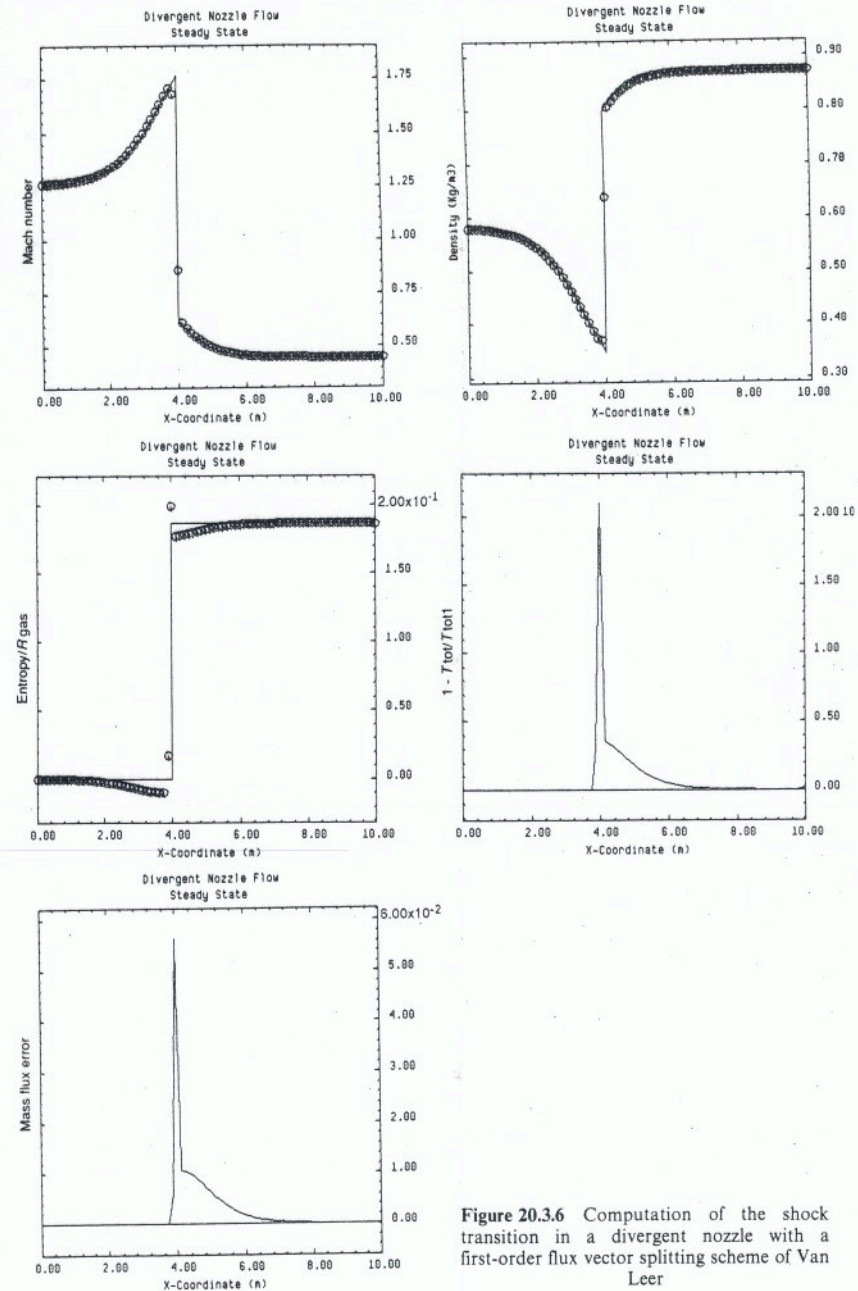


Figure 20.3.6 Computation of the shock transition in a divergent nozzle with a first-order flux vector splitting scheme of Van Leer

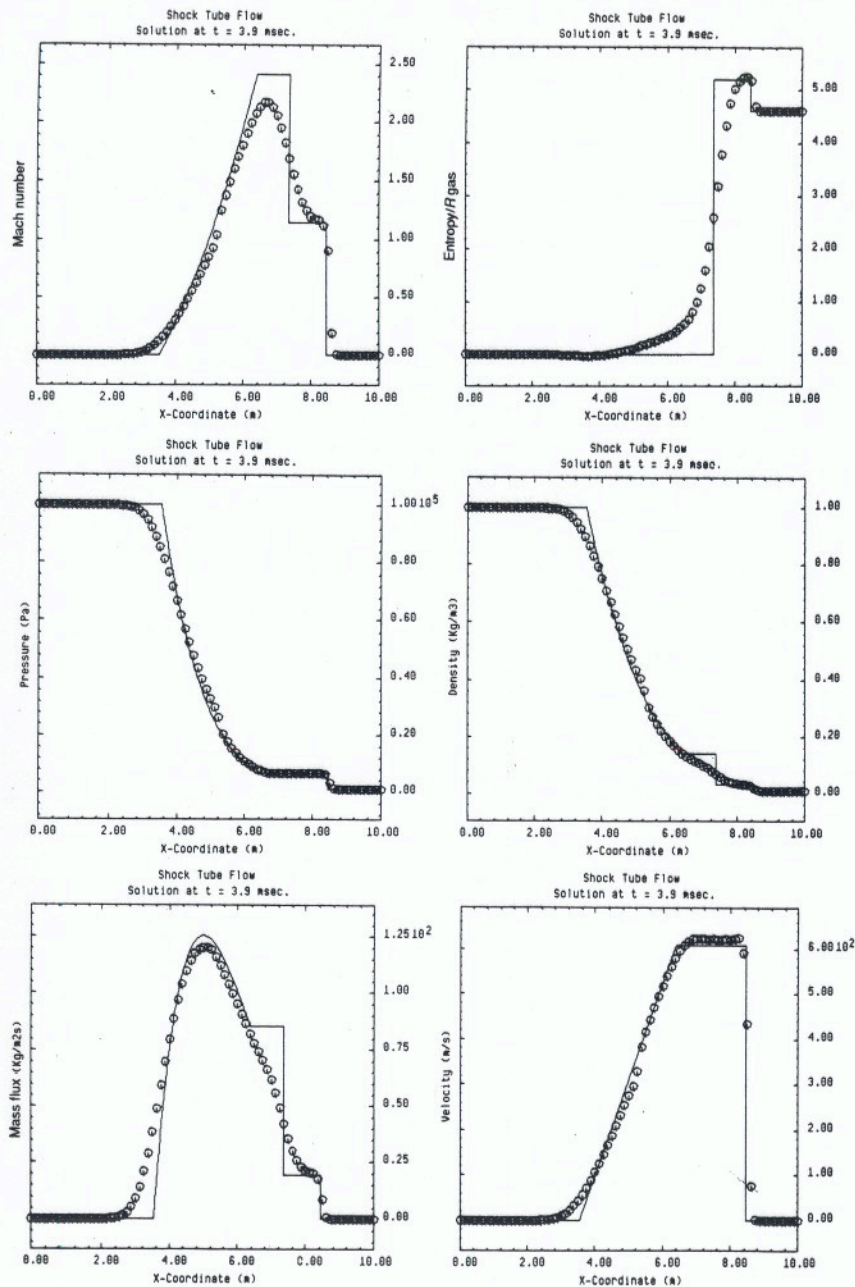


Figure 20.3.7 Computation of the shock tube problem of Figure 16.6.9 with a first-order scheme and the flux vector splitting of Steger and Warming, with $\varepsilon = 0$

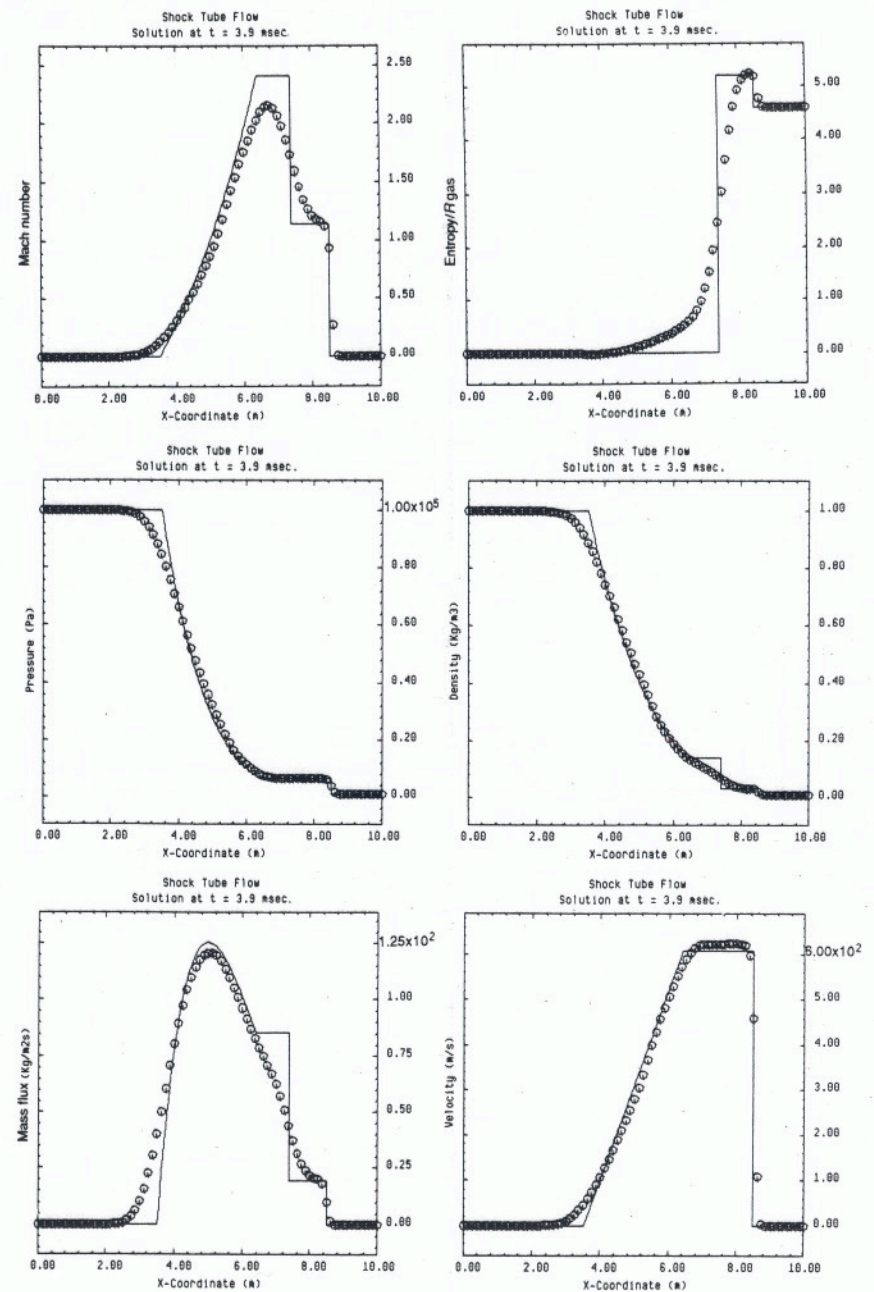


Figure 20.3.8 Computation of the shock tube problem of Figure 16.6.9 with a first-order scheme and the flux vector splitting of Steger and Warming, with $\varepsilon \neq 0$